Stat 6410 Exam 2 Review

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I Notation and the map of convergence modes

$(\Omega, \mathcal{F}, \mu)$ is a measure space. On a probability space $(\Omega, \mathcal{F}, P)$, a.e. reads a.s., in measure reads in probability, and $\int X\,dP = EX$. $f^+ = fI_{\{f\geq 0\}}$, $f^- = -fI_{\{f<0\}}$, $|f| = f^+ + f^-$, and $\int_A f\,d\mu \equiv \int fI_A\,d\mu$. $\|f\|_p = (\int |f|^p\,d\mu)^{\min\{1/p,1\}}$ for $0 < p < \infty$ (5.4), $\|f\|_\infty = \inf\{K : \mu(|f| > K) = 0\}$ (5.7), and $L^p(\mu) = \{f : \|f\|_p < \infty\}$ (Def. 2.3.7). $a_\lambda(t) = \int_{\{|f_\lambda|>t\}} |f_\lambda|\,d\mu$ (5.9). μT^{-1}: induced measure; P_X, F_X: law and cdf of X. $\mu \ll \nu$, $\mu \perp \nu$, $d\mu/d\nu$: absolutely continuous, singular, Radon–Nikodym derivative; $|\nu|$, $\nu^\pm$: total, positive and negative variation. φ, Φ: the $N(0,1)$ pdf and cdf; m: Lebesgue measure. In $[0,\infty]$: $0 \cdot \infty = 0$, and $\int f\,d\mu = \int f^+\,d\mu - \int f^-\,d\mu$ needs one of the two to be finite (Def. 2.3.4).

Mode	Definition	Consequences and cautions
pointwise (Def. 2.5.1)	$f_n(\omega) \to f(\omega)$ for every ω	gives a.e. with $B = \emptyset$
a.e. (Def. 2.5.2)	$\mu(B) = 0$ and $f_n \to f$ on B^c	passes through continuous g: $g(f_n) \to g(f)$ a.e.
in measure (Def. 2.5.3)	$\mu(\|f_n - f\| > \epsilon) \to 0$ for every $\epsilon > 0$	a.e. along a subsequence (Thm. 2.5.2)
L^p, $0 < p < \infty$ (Def. 2.5.4)	$f_n, f \in L^p$ and $\int \|f_n - f\|^p\,d\mu \to 0$	implies in measure (Thm. 2.5.3)
uniform (Def. 2.5.5)	$\sup_\Omega \|f_n - f\| \to 0$	with $\mu(\Omega) < \infty$, implies all the others (for L^p, given $f_n \in L^p$)
L^∞ (Def. 2.5.6)	$\|f_n - f\|_\infty \to 0$	uniform off a single null set
nearly uniform (Def. 2.5.7)	for each $\epsilon > 0$, uniform off some A with $\mu(A) < \epsilon$	"almost uniform"; implies a.e. and in measure

$$\text{uniform} \Rightarrow L^\infty \Rightarrow \text{nearly uniform} \Rightarrow \text{a.e.}, \qquad \text{nearly uniform} \Rightarrow \text{in measure} \Leftarrow L^p.$$

- If $\mu(\Omega) < \infty$: a.e. $\Rightarrow$ nearly uniform (Egorov, Thm. 2.5.11), hence a.e. $\Rightarrow$ in measure, which is Thm. 2.5.1 (Problem 2.26). Also $L^p \Rightarrow L^q$ for $0 < q < p$ (Problem 3.17; E9).
- Partial converses: in measure $\Rightarrow$ a.e. along a subsequence (Thm. 2.5.2); a.e. or in measure, plus $|f_n| \leq g \in L^1 \Rightarrow L^1$ (DCT, Cor. 2.3.12; Remark 2.5.1; E6); a.e. + UI + $\mu(\Omega) < \infty \Rightarrow L^1$ (Thm. 2.5.10); in measure + $\{|f_n|^p\}$ UI + $\mu(\Omega) < \infty \Rightarrow L^p$ (Problem 2.36(b)).
- $L^1 \iff$ in measure and $\int |f_n| \to \int |f|$ (Problem 2.29(a)); when $\mu(\Omega) < \infty$, also $\iff$ in measure and UI (Problem 2.29(b)).
- None of the unconditional arrows reverses, and no finiteness hypothesis can be dropped; Section VIII has a counterexample for each.

II Measurable functions and induced measures

Let $(\Omega, \mathcal{F})$ and $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2, 3$, be measurable spaces.

- **Measurable map** (Def. 2.1.3): $T : \Omega_1 \to \Omega_2$ is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$-measurable if $T^{-1}(A) \in \mathcal{F}_1$ for all $A \in \mathcal{F}_2$.
- **Measurable function; random variable** (Def. 2.1.2): $f : \Omega \to \mathbb{R}$ is $\mathcal{F}$-measurable if $\{f \leq a\} \in \mathcal{F}$ for every $a \in \mathbb{R}$ (1.1); on a probability space such an X is a random variable. Equivalently, $f^{-1}(A) \in \mathcal{F}$ for every $A \in \mathcal{B}(\mathbb{R})$.

Proposition 2.1.1. *(i) If $\mathcal{F}_2 = \sigma\langle\mathcal{A}\rangle$ and $T^{-1}(A) \in \mathcal{F}_1$ for all $A \in \mathcal{A}$, then T is $\langle\mathcal{F}_1, \mathcal{F}_2\rangle$-measurable. (ii) If T_1 is $\langle\mathcal{F}_1, \mathcal{F}_2\rangle$-measurable and T_2 is $\langle\mathcal{F}_2, \mathcal{F}_3\rangle$-measurable, then $T_2 \circ T_1$ is $\langle\mathcal{F}_1, \mathcal{F}_3\rangle$-measurable.*

Idea. (i) The good sets $\{A \in \mathcal{F}_2 : T^{-1}(A) \in \mathcal{F}_1\}$ form a σ-algebra containing $\mathcal{A}$, because T^{-1} commutes with complements and unions (Problem 2.1). (ii) $(T_2 \circ T_1)^{-1}(A) = T_1^{-1}(T_2^{-1}(A))$.

Apply. to show measurability, check a generating class: $\{f \le a\}$, $\{f < a\}$, or preimages of open sets.

Fact (building measurable functions). Continuous maps $\mathbb{R}^p \to \mathbb{R}^k$ are Borel measurable (Prop. 2.1.2). If $f_1, \dots, f_k$ are measurable, so are $(f_1, \dots, f_k)$, $\sum f_i$, $\prod f_i$, and $\psi(f_1, \dots, f_k)$ for continuous ψ (Prop. 2.1.3); measurable functions are closed under sums, products and scalar multiples (Cor. 2.1.4). $\sup_n f_n$, $\inf_n f_n$, $\limsup f_n$ and $\liminf f_n$ are measurable, and $\{\lim f_n \text{ exists and is finite}\} \in \mathcal{F}$ (Prop. 2.1.5). So are $(f/g)I(g \neq 0)$ (Problem 2.3) and $\sum_i I_{A_i} g_i$ for disjoint A_i (Problem 2.2(a)).

Apply. 2020 #2(b) (Y_i and $\hat{\beta}_n$; E8), 6410 #1(e) ($\cos f_n$), and every "argue that X is a random variable" step. Homework analogues: Problems 2.2, 2.3, 2.6 (roots of a random quadratic), 2.7 (determinant, trace, largest eigenvalue).

Definition 2.1.4 (generated σ-algebra). $\sigma\langle\{f_\lambda\}\rangle = \sigma\langle\{f_\lambda^{-1}(A) : A \in \mathcal{F}_2, \lambda \in \Lambda\}\rangle$, the smallest σ-algebra making every f_λ measurable. Ex. 2.1.3: $\sigma\langle I_A\rangle = \{\emptyset, A, A^c, \Omega\}$, so I_A is $\mathcal{F}$-measurable iff $A \in \mathcal{F}$. Ex. 2.1.4: the coordinate maps generate $\mathcal{B}(\mathbb{R}^k)$.

Apply. non-measurable examples (2020 #1(c)–(d); E7).

Proposition 2.2.1 (induced measure). *If T is $\langle\mathcal{F}_1, \mathcal{F}_2\rangle$-measurable and μ is a measure on $\mathcal{F}_1$, then $\mu T^{-1}(A) \equiv \mu(T^{-1}(A))$ is a measure on $\mathcal{F}_2$* (Problem 2.9). The law of X is $P_X = PX^{-1}$ (Def. 2.2.2), and its cdf is $F_X(x) = P_X((-\infty, x])$ (Def. 2.2.3).

Proposition 2.2.2, Definition 2.2.4 (cdf). F_X is nondecreasing and right-continuous, with $F(-\infty) = 0$ and $F(\infty) = 1$ (monotonicity, m.c.f.a., m.c.f.b.; Props. 1.2.1, 1.2.3). Any $F : \mathbb{R} \to \mathbb{R}$ with these three properties is called a cdf.

Fact (every cdf is the cdf of a random variable; §2.2). Given a cdf F, Thm. 1.3.3 and Def. 1.3.7 give the Lebesgue–Stieltjes probability measure μ_F on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with $\mu_F((-\infty, x]) = F(x)$; then $X(x) \equiv x$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F)$ has $P_X = \mu_F$ and $F_X = F$. Alternatively, on $((0,1), \mathcal{B}((0,1)), m)$ take $X(u) = F_1^{-1}(u) = \inf\{y : F(y) \ge u\}$ (Problem 2.15(c)).

Apply. the "exhibit a random variable" question (Section VII.2).

Fact (atoms and types). $P(X < x) = F(x-)$ and $P(X = x) = F(x) - F(x-)$ (2.4; Problem 2.10). X is discrete if $P(X \in A) = 1$ for some countable A, and continuous if $P(X = x) = 0$ for every x (Def. 2.2.5). Every cdf is $F = \alpha F_d + (1-\alpha)F_c$ with F_d discrete and F_c continuous (2.5); §4.5 splits F_c further. Homework analogues: Problem 2.14 (the measure that m induces under $x \mapsto x^2$), Problem 2.15 (cdfs of $\sin 2\pi x$ and $\log x$ under m on $(0,1)$).

III The integral and the convergence theorems

The integral in four steps (§2.3)

- **Simple** (Defs. 2.3.1–2.3.2): for $f = \sum_{i=1}^{k} c_i I_{A_i}$ with $c_i \in [0,\infty]$, $\int f\,d\mu = \sum_i c_i \mu(A_i)$; the value does not depend on the representation (Problem 2.17).
- **Nonnegative** (Def. 2.3.3): $\int f\,d\mu = \lim_n \int f_n\,d\mu$ for any simple $0 \le f_n \uparrow f$ (well defined by Prop. 2.3.2), e.g. f_n from (3.7) with $\delta_n = 2^{-n}$, $N_n = n2^n$. Equivalently $\sup\{\int g\,d\mu : g \text{ simple}, 0 \le g \le f\}$ (Remark 2.3.3).
- **General** (Defs. 2.3.4–2.3.5): $\int f\,d\mu = \int f^+\,d\mu - \int f^-\,d\mu$ when one of them is finite; f is integrable, $f \in L^1(\mu)$, if $\int |f|\,d\mu < \infty$.
- **Over a set** (Def. 2.3.6): $\int_A f\,d\mu \equiv \int f I_A\,d\mu$.

Proposition 2.3.1, Corollary 2.3.3, Proposition 2.3.8. *For nonnegative simple, then nonnegative measurable (Cor. 2.3.3, with $\alpha,\beta \ge 0$), then integrable f, g (Prop. 2.3.8, with $\alpha,\beta \in \mathbb{R}$): (i) $\int(\alpha f + \beta g) = \alpha \int f + \beta \int g$; (ii) $f \ge g$ a.e. $\Rightarrow \int f \ge \int g$; (iii) $f = g$ a.e. $\Rightarrow \int f = \int g$.*

Proposition 2.3.9. *If $f \ge 0$ a.e., then $\int f\,d\mu = 0$ iff $f = 0$ a.e.* **Proposition 2.3.10.** *If $f \in L^1$, then $|f| < \infty$ a.e.*

Idea. (2.3.9) $\mu(f > 1/n) \le n \int f\,d\mu = 0$, and $\{f > 1/n\} \uparrow \{f > 0\}$ (m.c.f.b.). (2.3.10) $\mu(|f| > n) \le n^{-1} \int |f|\,d\mu \to 0$, and $\{|f| > n\} \downarrow \{|f| = \infty\}$ (m.c.f.a.).

Apply. "equal integrals over every set ⇒ equal a.e." (E3), and the null-set step in Thm. 2.5.2.

Theorem 2.3.4 (monotone convergence theorem, MCT). *If $0 \le f_n \uparrow f$ a.e., then $\int f_n\,d\mu \uparrow \int f\,d\mu$.*

Corollary 2.3.5. *$\int \sum_n h_n\,d\mu = \sum_n \int h_n\,d\mu$ for nonnegative measurable h_n.* **Corollary 2.3.6.** *If $f \ge 0$ is measurable, $\nu(A) \equiv \int_A f\,d\mu$ is a measure,* and $\mu(A) = 0 \Rightarrow \nu(A) = 0$ (Remark 2.3.5).

Theorem 2.3.7 (Fatou's lemma). *For nonnegative measurable f_n, $\int \liminf_n f_n\,d\mu \le \liminf_n \int f_n\,d\mu$.*

Idea. $g_n = \inf_{j \ge n} f_j \uparrow \liminf f_n$; apply the MCT to g_n and use $g_n \le f_n$. The inequality can be strict: $I_{[n,\infty)}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$, and $nI_{[0,1/n]}$ on $[0,1]$ (Remark 2.3.6).

Corollary 2.3.12 (dominated convergence theorem, DCT). *If $|f_n| \le g$ a.e. for all n, $\int g\,d\mu < \infty$, and $f_n \to f$ a.e., then $f \in L^1$, $\int f_n\,d\mu \to \int f\,d\mu$, and $\int |f_n - f|\,d\mu \to 0$.*
Theorem 2.3.11 (extended DCT, EDCT). *The same conclusion holds if instead $|f_n| \le g_n$ a.e., $g_n \to g$ and $f_n \to f$ a.e., $g_n, g \in L^1$, and $\int g_n\,d\mu \to \int g\,d\mu$.*
Corollary 2.3.13 (bounded convergence theorem, BCT). *If $\mu(\Omega) < \infty$, $|f_n| \le k$ a.e., and $f_n \to f$ a.e., the same conclusion holds.*

Idea (EDCT). Fatou applied to the nonnegative $g_n + f_n$ and $g_n - f_n$ bounds $\liminf \int f_n$ from below and $\limsup \int f_n$ from above by $\int f$; apply the result again to $|f_n - f| \le g_n + |f|$.

Note. All three remain valid with "a.e." replaced by "in measure" (Remark 2.5.1; Problem 2.37; E6). Homework analogues: Problems 2.22 (continuous-parameter MCT and DCT), 2.33 (differentiating under the integral), 2.34–2.35, 2.40–2.41, 2.47.

Fact (integrating against a density; Problem 2.39). If $0 \le g \in L^1(\mu)$ and $\nu_g(A) = \int_A g\,d\mu$, then for measurable h: $h \in L^1(\nu_g)$ iff $hg \in L^1(\mu)$, and $\int h\,d\nu_g = \int hg\,d\mu$.

Fact (change of variables; Problem 2.42, marked "His favorite" in your Chapter 2 notes). If $f : \Omega_1 \to \Omega_2$ is $\langle \mathcal{F}_1, \mathcal{F}_2\rangle$-measurable, $\mu_2 = \mu_1 f^{-1}$, and $h : \Omega_2 \to \mathbb{R}$ is $\langle \mathcal{F}_2, \mathcal{B}(\mathbb{R})\rangle$-measurable, then $h \circ f \in L^1(\mu_1)$ iff $h \in L^1(\mu_2)$ iff the identity map is in $L^1(\mu_3)$, $\mu_3 = \mu_2 h^{-1}$, and

$$\int_{\Omega_1} h \circ f\,d\mu_1 = \int_{\Omega_2} h\,d\mu_2 = \int_{\mathbb{R}} x\,d\mu_3(x).$$

In probability: $Eh(X) = \int h\,dP_X$ (E14).

Technique (the standard machine). To prove an identity between integrals, check it for indicators, extend to nonnegative simple functions by linearity, to nonnegative measurable functions by the MCT, and to integrable functions through $f = f^+ - f^-$. This is how Problems 2.39 and 2.42 and Prop. 4.1.2 are proved.

Theorem 2.4.1 (Riemann versus Lebesgue). *A bounded f on $[a, b]$ is Riemann integrable iff it is continuous a.e. (m); then the Lebesgue and Riemann integrals agree.* $I_{\mathbb{Q}\cap[0,1]}$ is Lebesgue but not Riemann integrable (Problem 2.25). For improper integrals, integrate over $[-n, n]$ and let $n \to \infty$ with the MCT or DCT (Problem 2.31).

Apply. every "give a numerical answer" step, e.g. $\int x\varphi(x+1)\,dx = -1$ (E5).

Also in §2.3 (statement only; marked "skip" in your notes). Thm. 2.3.14: step functions, and continuous functions with compact support, are dense in $L^p(\mu_F)$ for $0 < p < \infty$; this fails for $p = \infty$ (Remark 2.3.10).

IV More on convergence and uniform integrability

Theorem 2.5.1. *If $\mu(\Omega) < \infty$, then $f_n \to f$ a.e. implies $f_n \to f$ in measure.*

Idea (Problem 2.26). With $A_{jr} = \{|f_j - f| > 1/r\}$, the set where $f_n \not\to f$ is $D = \bigcup_r \bigcap_n \bigcup_{j\ge n} A_{jr}$. Since $\mu(D) = 0$ and $\mu(\Omega) < \infty$, m.c.f.a. gives $\mu(A_{nr}) \le \mu(\bigcup_{j\ge n} A_{jr}) \downarrow 0$ for each r. Finiteness is needed: $I_{[n,\infty)} \to 0$ everywhere on $(\mathbb{R}, m)$, but not in measure.

Theorem 2.5.2. *If $f_n \to f$ in measure, then $f_{n_k} \to f$ a.e. for some subsequence.*

Idea. Choose $n_k \uparrow$ with $\mu(A_k) < 2^{-k}$, $A_k = \{|f_{n_k} - f| > 2^{-k}\}$. Then $\int \sum_k I_{A_k} = \sum_k \mu(A_k) < \infty$ (Cor. 2.3.5), so $\sum_k I_{A_k} < \infty$ a.e. (Prop. 2.3.10): a.e. ω lies in only finitely many A_k. A subsequence is all one gets: the typewriter sequence converges nowhere (Problem 2.46; E16).

Theorem 2.5.3. *If $f_n \to f$ in L^p for some $0 < p < \infty$, then $f_n \to f$ in measure.*

Idea. Markov: $\mu(|f_n - f| \ge \epsilon) \le \epsilon^{-p} \int |f_n - f|^p\,d\mu$. The converse fails: $nI_{[0,1/n]} \to 0$ in measure on $[0, 1]$, yet $\int |f_n|\,dm = 1$ for every n.

Theorem 2.5.4 (Scheffé). *If $f_n, f \ge 0$, $f_n \to f$ a.e., and $\int f_n\,d\mu \to \int f\,d\mu < \infty$, then $\int |f_n - f|\,d\mu \to 0$.* Cor. 2.5.5: pdfs that converge a.e. converge in L^1; Cor. 2.5.6: the same for pmfs; Remark 2.5.2: in measure suffices.

Idea. $0 \le (f-f_n)^+ \le f$, so the DCT gives $\int (f-f_n)^+ \to 0$; then $\int (f-f_n)^- = \int (f-f_n)^+ - \int (f-f_n) \to 0$.

Apply. Binomial$(n,p_n) \to$ Poisson(λ) in total variation iff $np_n \to \lambda$ (Problem 4.7(c)); normal densities (Problem 2.43); Problem 2.44.

Definition 2.5.8 (uniform integrability). $\{f_\lambda : \lambda \in \Lambda\} \subset L^1$ *is UI if* $\sup_\lambda a_\lambda(t) \to 0$ *as* $t \to \infty$, *where* $a_\lambda(t) = \int_{\{|f_\lambda|>t\}} |f_\lambda|\, d\mu$ (5.9)–(5.10).

Proposition 2.5.7. *(i) A finite family in* L^1 *is UI. (ii) If* $\sup_\lambda \int |f_\lambda|^{1+\epsilon}\, d\mu < \infty$ *for some* $\epsilon > 0$, *the family is UI. (iii) If* $|f_\lambda| \le g$ *a.e. with* $g \in L^1$, *the family is UI. (iv) If* $\{f_\lambda\}$ *and* $\{g_\gamma\}$ *are UI, so is* $\{f_\lambda + g_\gamma\}$. *(v) If* $\{f_\lambda\}$ *is UI and* $\mu(\Omega) < \infty$, *then* $\sup_\lambda \int |f_\lambda|\, d\mu < \infty$.

More generally, $\sup_\lambda \int \phi(|f_\lambda|)\, d\mu < \infty$ with $\phi(x)/x \uparrow \infty$ gives UI (Remark 2.5.3; Problem 2.27). L^1-boundedness alone does not: $nI_{[0,1/n]}$.

Proposition 2.5.8 (absolute continuity of the integral). *If* $f \in L^1$, *every* $\epsilon > 0$ *has a* $\delta > 0$ *with* $\mu(A) < \delta \Rightarrow \int_A |f|\, d\mu < \epsilon$. **Theorem 2.5.9.** *If* $\mu(\Omega) < \infty$, *a family in* L^1 *is UI iff it is* L^1*-bounded and uniformly absolutely continuous* (Def. 2.5.9).

Theorem 2.5.10 (UI convergence theorem). *Let* $\mu(\Omega) < \infty$, *let* $\{f_n\} \subset L^1$ *be UI, and let* $f_n \to f$ *a.e. with* f *measurable. Then* $f \in L^1$ *and* $\int |f_n - f|\, d\mu \to 0$.

Remark 2.5.4. For $f_n \ge 0$ with $f_n \to f$ a.e. and $\mu(\Omega) < \infty$: $\int f_n\, d\mu \to \int f\, d\mu < \infty$ iff $\{f_n\}$ is UI (Problem 2.29). UI is sufficient but not necessary for $\int f_n \to \int f$ (Problem 2.28), and it is not sufficient when $\mu(\Omega) = \infty$ (Problem 2.30).

Apply. the one-line "not UI" argument for the exam's spike sequences: they converge a.e. on a finite measure space but not in L^1, so Thm. 2.5.10 rules out UI (Section VII.1).

Also in §2.5 (statements only). Thm. 2.5.11 (Egorov): if $\mu(\Omega) < \infty$ and $f_n \to f$ a.e., then $f_n \to f$ nearly uniformly. Thm. 2.5.12 (Lusin): an a.e.-finite μ_F^*-measurable f equals a continuous function off a set of arbitrarily small measure. With Thm. 1.3.4 these are Littlewood's three principles (Remark 2.5.5). Homework analogues for this section: Problems 2.26–2.30, 2.36, 2.37, 2.43–2.47.

V Inequalities and L^p spaces

Theorem 3.1.1 (Markov's inequality). *If* $f \ge 0$ *is measurable and* $0 < t < \infty$, *then* $\mu(f \ge t) \le t^{-1} \int f\, d\mu$.

Corollaries 3.1.2–3.1.5. $P(|X| \ge t) \le E|X|^r/t^r$; Chebyshev, $P(|X - EX| \ge k\sigma) \le 1/k^2$; $P(|X| \ge t) \le E\phi(|X|)/\phi(t)$ for nondecreasing $\phi \ge 0$; Cramér, $P(X \ge t) \le \inf_{\theta>0} Ee^{\theta X}/e^{\theta t}$.

Apply. $L^p \Rightarrow$ in measure (Thm. 2.5.3), the completeness proof (Thm. 3.2.2), and Thm. 2.5.9.

Definition 3.1.1 (convex function). $\varphi : (a,b) \to \mathbb{R}$ *is convex if* $\varphi(\lambda x + (1-\lambda)y) \le \lambda\varphi(x) + (1-\lambda)\varphi(y)$ *for all* $x, y \in (a,b)$ *and* $\lambda \in [0,1]$.

Propositions 3.1.6–3.1.8. Chord slopes increase (1.3)–(1.4); the one-sided derivatives $\varphi'_\pm$ exist, are nondecreasing, and give supporting lines $\varphi(x) - \varphi(c) \ge \varphi'_+(c)(x-c)$ (1.5); if $\varphi'' \ge 0$ on (a,b), then φ is convex (Prop. 3.1.8). Convex (Ex. 3.1.1): $|x|^p$ $(p \ge 1)$ and e^x on $\mathbb{R}$; $-\log x$ and $x \log x$ on $(0,\infty)$. Also $\sqrt{1+x^2}$ on $\mathbb{R}$ and $1/x$ on $(0,\infty)$.

Theorem 3.1.9 (Jensen's inequality). *Let f be measurable on a probability space $(\Omega, \mathcal{F}, P)$ with $P(f \in (a,b)) = 1$, $-\infty \le a < b \le \infty$, and let φ be convex on (a,b). If $\int |f|\,dP < \infty$ and $\int |\varphi(f)|\,dP < \infty$, then $\varphi\big(\int f\,dP\big) \le \int \varphi(f)\,dP$,* i.e. $\varphi(EX) \le E\varphi(X)$ (Remark 3.1.2).

Idea. The supporting line at $c = EX$: $\varphi(X) - \varphi(c) - \varphi'_+(c)(X - c) \ge 0$ (1.5); take expectations. For strictly convex φ, equality holds iff $X = c$ a.s. (Remark 3.1.3).

Note (hypotheses). P must be a probability measure, and both integrability conditions must be checked, often with an elementary bound such as $\sqrt{1+x^2} \le 1 + x$. When $EX = \infty$, argue separately (E10).

Apply. 6410 #3(a) (E10), 2020 #3(iv) (E9), and Problems 3.5–3.8, 3.17, 3.18 (E12).

Proposition 3.1.10. (ii) $\sum p_i b_i \ge \prod b_i^{p_i}$ for $b_i \ge 0$, $p_i > 0$, $\sum p_i = 1$ (1.10), so the arithmetic mean dominates the geometric mean (1.11); (iii) $|a+b|^p \le 2^{p-1}(|a|^p + |b|^p)$ for $1 \le p < \infty$ (1.12). For $0 < p \le 1$ and $x, y \ge 0$, $(x+y)^p \le x^p + y^p$ (2.2).

Theorem 3.1.11 (Hölder's inequality). *Let $1 < p < \infty$ and $q = p/(p-1)$. If $f \in L^p$ and $g \in L^q$, then $\int |fg|\,d\mu \le \big(\int |f|^p\big)^{1/p}\big(\int |g|^q\big)^{1/q}$; if $\|fg\|_1 \ne 0$, equality holds iff $|f|^p = c|g|^q$ a.e.* The case $p = 1$, $q = \infty$ also holds (Remark 3.1.4).

Corollary 3.1.12 (Cauchy–Schwarz). $\int |fg|\,d\mu \le \big(\int |f|^2\big)^{1/2}\big(\int |g|^2\big)^{1/2}$.

Theorem 3.1.14 (Minkowski). $\|f+g\|_p \le \|f\|_p + \|g\|_p$ *for $f, g \in L^p$, $1 < p < \infty$;* also for $p = 1, \infty$ (Remark 3.1.5).

Idea (Hölder). Apply (1.10) with $k = 2$ at each ω to $|f|^p/\int |f|^p$ and $|g|^q/\int |g|^q$ (Young's inequality), then integrate.

Apply. 2019 #2(b) (E2), Problems 3.8–3.10, and every moment comparison.

Fact (moments on finite measure spaces; Problems 3.16–3.17). If $\mu(\Omega) < \infty$ and $0 < q < p < \infty$, Hölder with exponent p/q applied to $|f|^q \cdot 1$ gives

$$\int |f|^q\,d\mu \le \Big(\int |f|^p\,d\mu\Big)^{q/p} \mu(\Omega)^{1-q/p}.$$

So $L^p \subset L^q$, and L^p convergence implies L^q convergence (E9). On a probability space this is Lyapunov's inequality, $(E|X|^q)^{1/q} \le (E|X|^p)^{1/p}$ (Problem 3.17(a)). Without $\mu(\Omega) < \infty$, neither inclusion holds (Section VIII).

Proposition 3.2.1, Definitions 3.2.1–3.2.2. $L^p(\mu)$ is a vector space for $0 < p \le \infty$ (via (2.2) for $p \le 1$ and (1.12) for $p > 1$). Identifying $f \sim g$ when $f = g$ a.e., $d_p(f,g) = \|f - g\|_p$ is a metric on equivalence classes. For $0 < p < 1$, $d_p(f,g) = \int |f-g|^p\,d\mu$ is the metric (Remark 3.2.1), since $(\int |f|^p)^{1/p}$ fails the triangle inequality (Problem 3.12(a)).

Theorem 3.2.2 (completeness). $(L^p(\mu), d_p)$ *is complete for $0 < p \le \infty$.*

Idea. Pick $n_k \uparrow$ with $\int |f_n - f_m|^p \le 2^{-(p+1)k}$ for $n, m \ge n_k$. Markov and a Borel–Cantelli step (Problem 3.13) make f_{n_k} converge a.e. to some f; Fatou puts f in L^p with $d_p(f_{n_k}, f) \to 0$; the triangle inequality finishes.

Theorem 3.2.3 (Riesz representation). *For $1 \le p < \infty$, every continuous linear functional T on $L^p(\mu)$ has the form $T(f) = \int fg\,d\mu$ for some $g \in L^q$, with $q = p/(p-1)$ ($q = \infty$ if $p = 1$).* It fails for $p = \infty$ (Remark 3.2.2). Problem 4.30 derives it from the Radon–Nikodym theorem.

Also in §3.3 (statements only). A Banach space is a complete normed space (Def. 3.3.2), e.g. L^p for $p \geq 1$, and $C[0,1]$ with the sup norm (Problem 3.22). A Hilbert space is a complete inner-product space (Def. 3.3.7), e.g. $L^2(\mu)$ with $\langle f, g\rangle = \int fg\, d\mu$. Bessel's inequality (Prop. 3.3.2) and the Riesz representation for Hilbert spaces (Thm. 3.3.3) are the tools behind the proof of the Radon–Nikodym theorem (Section VI).

VI Radon–Nikodym, decompositions and signed measures

Let μ, ν be measures on a measurable space $(\Omega, \mathcal{F})$.

- **Absolute continuity** (Def. 4.1.1): $\mu \ll \nu$ if $\nu(A) = 0 \Rightarrow \mu(A) = 0$ for every $A \in \mathcal{F}$.
- **Singularity** (Def. 4.1.2): $\mu \perp \nu$ if $\mu(B) = 0$ and $\nu(B^c) = 0$ for some $B \in \mathcal{F}$. Then $\mu(A) = \mu(A \cap B^c)$ and $\nu(A) = \nu(A \cap B)$ for every A (1.4). $\perp$ is symmetric; $\ll$ is not.

Examples (Exs. 4.1.1–4.1.5).

- $N(0,1) \ll m$ (Ex. 4.1.1). In general, $\mu(A) = \int_A f\, d\nu$ with $f \geq 0$ gives $\mu \ll \nu$ (Ex. 4.1.3).
- Poisson(λ) and counting measure on $\mathbb{Z}_+$ are mutually absolutely continuous: both vanish only on $\emptyset$ (Ex. 4.1.2).
- $m \perp$ counting measure on $\mathbb{Z}$, with $B = \mathbb{Z}$ (Ex. 4.1.4); m restricted to $(-\infty, 0] \perp$ Exponential(1), with $B = (-\infty, 0]$ (Ex. 4.1.5); $m \perp$ the Cantor measure (§4.5).

Theorem 4.1.1 (Lebesgue decomposition and Radon–Nikodym). *Let μ_1, μ_2 be σ-finite measures on $(\Omega, \mathcal{F})$. (i) $\mu_1 = \mu_{1a} + \mu_{1s}$ for unique σ-finite measures with $\mu_{1a} \ll \mu_2$ and $\mu_{1s} \perp \mu_2$. (ii) There is a measurable $h \geq 0$ with $\mu_{1a}(A) = \int_A h\, d\mu_2$ for every $A \in \mathcal{F}$.*

Idea. For finite measures, $f \mapsto \int f\, d\mu_1$ is a bounded linear functional on $L^2(\mu_1 + \mu_2)$ (Cauchy–Schwarz), so Riesz (Thm. 3.3.3) gives g with $\int f\, d\mu_1 = \int fg\, d(\mu_1 + \mu_2)$ and $0 \leq g \leq 1$ a.e. (Problem 4.1). Then μ_{1s} lives on $\{g = 1\}$, and $h = g/(1-g)$ on $\{g < 1\}$. For σ-finite measures, work on a partition of Ω into pieces of finite measure (Problem 4.2).

Remark 4.1.1 (σ-finiteness matters). Lebesgue measure on $[0,1]$ is $\ll$ counting measure ν, but no h has $m(A) = \int_A h\, d\nu$ for all A: such an h would be positive only on a countable set (Problem 4.3).

Definition 4.1.3 (Radon–Nikodym derivative). *If $h \geq 0$ is measurable and $\mu(A) = \int_A h\, d\nu$ for all $A \in \mathcal{F}$, then $d\mu/d\nu = h$.* It is unique a.e. (ν) when μ is σ-finite (E3).

Proposition 4.1.2 (calculus of RN derivatives; all measures σ-finite). *(i) Chain rule: if $\mu_1 \ll \mu_2 \ll \mu_3$, then $\mu_1 \ll \mu_3$ and $\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3}$ a.e. (μ_3). (ii) Linearity: if $\mu_1, \mu_2 \ll \mu_3$ and $\alpha, \beta \geq 0$, then $\frac{d(\alpha\mu_1 + \beta\mu_2)}{d\mu_3} = \alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}$ a.e. (μ_3). (iii) Reciprocal: if $\mu \ll \nu$ and $\frac{d\mu}{d\nu} > 0$ a.e. (ν), then $\nu \ll \mu$ and $\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1}$ a.e. (μ). (iv) Sums: if $\mu = \sum_n \alpha_n \mu_n$ with $\alpha_n > 0$, then $\mu \ll \nu$ iff every $\mu_n \ll \nu$, with $\frac{d\mu}{d\nu} = \sum_n \alpha_n \frac{d\mu_n}{d\nu}$; and $\mu \perp \nu$ iff every $\mu_n \perp \nu$.*

Idea. The standard machine with Problem 2.39 (Problem 4.4).

Apply. E5, E11, E17; Problems 4.5 and 4.13.

Fact (decomposing a mixture). If $\mu = \mu_c + \sum_j b_j \delta_{x_j}$ with $\mu_c(A) = \int_A f\, dm$, then with respect to m: $\mu_a = \mu_c$ with $d\mu_a/dm = f$, and $\mu_s = \sum_j b_j \delta_{x_j}$, with singular set $B = \{x_1, x_2, \dots\}$ (countable,

so $m(B) = 0$). Against a discrete ν that puts positive mass on each point of a countable set S and none off S, the roles swap: $\mu_a(A) = \mu(A \cap S)$, $\frac{d\mu_a}{d\nu}(y) = \mu(\{y\})/\nu(\{y\})$ for $y \in S$, and the rest of μ is singular (Problem 4.5(d), (f)).

Signed measures (§4.2). A finite signed measure (Def. 4.2.1) is a countably additive $\nu : \mathcal{F} \to \mathbb{R}$ with $\nu(\emptyset) = 0$ and finite total variation. Its total variation $|\nu|$ is a finite measure (Prop. 4.2.1, Def. 4.2.2), and $\nu^{\pm} = (|\nu| \pm \nu)/2$ (Def. 4.2.3).

Theorem 4.2.2. *ν is a finite signed measure iff $\nu = \mu_1 - \mu_2$ for finite measures μ_1, μ_2, iff $\nu(A) = \int_A f\, d\mu$ for a finite measure μ and some $f \in L^1(\mu)$.*
Theorem 4.2.3 (Hahn decomposition). *There are disjoint $\Omega^+, \Omega^- \in \mathcal{F}$ with $\Omega = \Omega^+ \cup \Omega^-$, where Ω^+ is a positive set and Ω^- a negative set for ν (Def. 4.2.4).*

Then $\nu^+(A) = \nu(A \cap \Omega^+)$ and $\nu^-(A) = -\nu(A \cap \Omega^-)$, and $\nu = \nu^+ - \nu^-$ is the Jordan decomposition into mutually singular measures. It is minimal: any $\nu = \mu_1 - \mu_2$ has $\mu_1 = \nu^+ + \lambda$, $\mu_2 = \nu^- + \lambda$ for a measure λ, with $\lambda = 0$ iff $\mu_1 \perp \mu_2$ (Thm. 4.2.4). $\|\nu\| = |\nu|(\Omega)$ is a complete norm (Def. 4.2.5, Props. 4.2.5–4.2.6). For $\nu_f(A) = \int_A f\, d\mu$: $|\nu_f|(A) = \int_A |f|\, d\mu$ and $\nu_f^{\pm} = \nu_{f^{\pm}}$ (Problem 4.6). For probability measures, $\|\mu_1 - \mu_2\| = 2 \sup_A |\mu_1(A) - \mu_2(A)|$ (Problem 4.10), and $|\mu_1 - \mu_2|(A) = \int_A |h - 1|\, d\mu_2 + \mu_{1s}(A)$ with $h = d\mu_{1a}/d\mu_2$ (Problem 4.13(a)).

Also in Chapter 4 (statements only; not yet on a past exam).

- **Bounded variation** (§4.3). $f \in BV[a, b]$ iff $T(f, [a, b]) < \infty$ (Def. 4.3.3), iff f is a difference of two nondecreasing functions (Thm. 4.3.1), iff $f(x) = \mu([a, x])$ for a finite signed μ (Cor. 4.3.2). Lipschitz functions and $x^2 \sin(1/x)$ are BV on $[0, 1]$; $x^2 \sin(1/x^2)$ and $I_{\mathbb{Q}}$ are not (Remark 4.3.2).
- **Absolutely continuous functions** (§4.4). F is a.c. on $[a, b]$ iff $F(x) = F(a) + \int_{[a,x]} f\, dm$ with $f \in L^1$ (Thm. 4.4.1), and then $F' = f$ a.e. (Thm. 4.4.2). A nondecreasing F is a.c. on every bounded interval iff $\mu_F \ll m$ (Thm. 4.4.3); then $\int h\, d\mu_F = \int hf\, dm$ (Problem 4.19).
- **Singular distributions** (§4.5). F is singular if $F' = 0$ a.e. (Def. 4.5.1). Every cdf is $F = \alpha F_d + \beta F_{ac} + \gamma F_{sc}$ (Prop. 4.5.1). The Cantor set C is uncountable with $m(C) = 0$ (Thm. 4.5.2, Cor. 4.5.3), and the Cantor function is a continuous singular cdf: $\mu_F \perp m$, so it is not a.c.

Homework analogues: Problems 4.1–4.6, 4.10, 4.13 (§§4.1–4.2); 4.15–4.22, 4.29 (§§4.3–4.4); 4.31–4.33 (§4.5).

VII Exam question bank

Scope. Chapters 2–4. Past exams: the 2019 and 2020 Midterm 2 (Stat 641, 100 points, with keys) and the scanned Stat 6410 Midterm 2 (110 points, max score 100), called "6410" below and solved in full in `stat6410_midterm2_solutions.tex`. Each exam has a convergence-modes question worth 40–50 points; the rest mixes the integration theorems, measurability and existence of random variables, inequalities, and Radon–Nikodym (2019 and 6410). The 2019 exam attached Tables 4.6.1–4.6.2 (standard discrete and absolutely continuous distributions) as its second page.

Ghosh's instructions. "Write legibly showing all the steps VERY clearly" (every exam); "You can state and use any calculus/basic statistics facts without proofs (but explain the *measure-theoretic* steps)" (2019 #3); "You can state & use any pre-measure theory facts you know (if

needed), but justify all the measure theory steps" (6410 #2); "you can state any result proved in class to argue this" (6410 #3(b)). So compute freely with calculus, but cite a numbered result at every measure-theoretic step.

Question type	Seen on	Analogous problems
Modes of convergence for a spike sequence (Section VII.1)	2019 #1; 2020 #3(i)–(iii); 6410 #1	2.26, 2.28, 2.30, 2.46; Remarks 2.3.6, 2.5.3
Moment comparison: $L^p \Rightarrow L^q$	2020 #3(iv)	3.16, 3.17
State or extend the DCT; Fatou variants	2019 #2(a); 2020 #1(a)–(b)	2.22, 2.37, 2.41
Measurability; non-measurable examples	2020 #1(c)–(d), #2(b)	2.2, 2.3, 2.6, 2.7
Exhibit a random variable with a given law (Section VII.2)	2020 #2(a); 6410 #2(i)	2.15; Problems 1.22–1.23
Compute an integral or a moment	2019 #3(c); 2020 #2(c); 6410 #2(ii)	2.14, 2.31, 2.39, 2.42
Jensen or Cauchy–Schwarz inequalities	2019 #2(b); 6410 #3(a)	3.5, 3.6, 3.8, 3.17, 3.18
Equal integrals on all sets $\Rightarrow$ equal a.e.	2019 #2(c)	2.45; Prop. 2.3.9
$\mu \ll \nu$ and $\mu \perp \nu$ together	2019 #3(a); 6410 #3(c)	Def. 4.1.2 and (1.4); 4.2
Lebesgue decomposition; RN derivatives	2019 #3(b); 6410 #3(b)	4.4, 4.5, 4.13; 2.39

Note (Ghosh reuses the homework). 2019 #2(b) is Problem 3.8(a) restated for a measure space, and 6410 #3(a) is Problem 3.18(b) word for word, plus an infinite-mean case. 2020 #1(b) is Problem 2.37(b) specialized to the DCT, and 2019 #3(b) is a Problem 4.5 decomposition. Section VII.5 works the closest unasked neighbours.

VII.1 The convergence-modes question (on every past exam)

Each part is worth 10 points, and a yes/no part still needs a proof. State the answer first, then argue from the definition or a numbered theorem.

Model answer: a spike $f_n = c_n I_{A_n} \to f \equiv 0$ with $\mu(A_n) \to 0$, $c_n \to \infty$, $\mu(\Omega) = 1$

(a) a.e. Fix ω. Show $\omega \notin A_n$ for all large n, for every ω off an explicit null set N (often $N = \emptyset$ or a single point), so $f_n(\omega) = 0$ eventually. Hence $f_n \to f$ on N^c, i.e. a.e. (Defs. 2.5.1–2.5.2). $\mu(A_n) \to 0$ alone is not enough (Problem 2.46).
(b) In measure. For $\epsilon > 0$, $\{|f_n - f| > \epsilon\} \subset A_n$, so $\mu(|f_n - f| > \epsilon) \le \mu(A_n) \to 0$ (Def. 2.5.3). Alternatively cite Thm. 2.5.1, since $\mu(\Omega) < \infty$ and (a) holds.
(c) L^p. $|f_n - f|^p = c_n^p I_{A_n}$ is simple, so $\int |f_n - f|^p \, d\mu = c_n^p \mu(A_n)$ (Def. 2.3.2). Find the threshold p^* beyond which $c_n^p \mu(A_n) \not\to 0$; then $f_n \to f$ in L^p iff $p < p^*$ (Def. 2.5.4). Check the boundary $p = p^*$ separately.
(d) UI. For $t > 0$ and every n with $c_n > t$, $\{|f_n| > t\} = A_n$, so $a_n(t) = c_n \mu(A_n)$. Hence $\sup_n a_n(t) \ge \limsup_n c_n \mu(A_n)$ for every t, and if $c_n \mu(A_n) \not\to 0$ the family is not UI (Def. 2.5.8). Shortcut: $f_n \ge 0$, $f_n \to f$ a.e. and $\mu(\Omega) < \infty$, so UI holds iff $\int f_n \to \int f$ (Remark 2.5.4; Thm. 2.5.10), i.e. iff (c) holds at $p = 1$.

(e) Transformations. If g is continuous, $g(f_n(\omega)) \to g(f(\omega))$ wherever $f_n(\omega) \to f(\omega)$, so a.e. convergence passes to $g(f_n)$.

	2019 #1	2020 #3	6410 #1
Sequence	$X_n = 1 + (n-1)I_{(0,1/n]}$ $(n \geq 2)$, $X \equiv 1$, on $(0,1)$	$X_n = n^2 I_{[1-1/n,\,1]}$, $X \equiv 0$, on $[0,1]$	$f_n = nI_{[1/n,\,2/n]}$, $f \equiv 0$, on $[0,1]$
a.e.	yes, everywhere: $X_n(\omega) = 1$ once $1/n < \omega$	yes, off $\{1\}$: $X_n(1) = n^2 \to \infty$	yes, everywhere: $f_n(x) = 0$ once $n > 2/x$, and $f_n(0) = 0$
in measure	yes: $P(\lvert X_n - X\rvert > \epsilon) \leq 1/n$	yes: $m([1-\frac{1}{n}, 1]) = \frac{1}{n}$	yes: $m(\lvert f_n\rvert > \epsilon) \leq \frac{1}{n}$
$\int \lvert f_n - f\rvert^p$	$(n-1)^p/n$	n^{2p-1}	n^{p-1} $(n \geq 2)$
L^p	iff $0 < p < 1$	iff $0 < p < \frac{1}{2}$	iff $0 < p < 1$
UI	no: $a_n(t) = 1$ for $n > t \geq 1$	no (not asked): $a_n(t) = n$ for $n^2 > t$	no: $a_n(t) = 1$ for $n > \max(t, 1)$
Extra	—	(iv) $L^p \Rightarrow L^q$, $q < p$ (E9)	(e) $\cos f_n \to 1$ everywhere

Note (slips to avoid). Name the exceptional null set when convergence fails somewhere: in 2020 #3, $X_n(1) = n^2$ (the key's "$X_n(1) = 1$" is a typo). At the L^p threshold the integral stays at 1 (2019 and 6410 at $p = 1$) or grows, so the boundary belongs to "does not converge." For UI, take the supremum over n before letting $t \to \infty$; a bound for each fixed n proves nothing.

VII.2 The "exhibit a random variable" question (2020 #2, 6410 #2)

2020 asked to "argue that there exists a random variable … whose distribution on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is Normal(0,1)"; 6410 asked, "using steps of the Carathéodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable X" with the Binomial$(10, 0.1)$ pmf. The second wants the full Exam 1 template, compressed below.

Model answer: a random variable with cdf F

Step 1. *cdf.* Verify that F is a cdf (Def. 2.2.4). Discrete: $F(x) = \sum_{k \leq x} p_k$ is a right-continuous step function with $F(\infty) = \sum_k p_k = 1$. Density: $F(x) = \int_{(-\infty, x]} f\, dm$ is continuous with $F(\infty) = 1$.
Step 2. *Semialgebra.* $\mathcal{C} = \{(a, b] : -\infty \leq a \leq b < \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\}$, with $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$ (Exam 1 Review, Section V).
Step 3. *Measure on $\mathcal{C}$.* $\mu_F((a, b]) = F(b) - F(a)$ and $\mu_F((a, \infty)) = 1 - F(a)$; countably additive by Heine–Borel (Problems 1.22–1.23).
Step 4. *Extension.* μ_F^* is an outer measure, $\mathcal{M}_{\mu_F^*} \supset \mathcal{C}$ is a σ-algebra on which μ_F^* is a measure, and $\mu_F^* = \mu_F$ on $\mathcal{C}$ (Thms. 1.3.2–1.3.3).
Step 5. *Restrict.* $P \equiv \mu_F^*|_{\mathcal{B}(\mathbb{R})}$ is a probability measure, $P(\mathbb{R}) = F(\infty) - F(-\infty) = 1$, unique by Thm. 1.2.4 (Def. 1.3.7).
Step 6. *Random variable.* On $(\Omega, \mathcal{F}, P) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), P)$ let $X(\omega) = \omega$. Then $X^{-1}((-\infty, a]) = (-\infty, a] \in \mathcal{B}(\mathbb{R})$ (Def. 2.1.2(b)), $P_X = P$ and $F_X = F$ (§2.2).
Step 7. *Check the law.* Discrete: $P(X = x) = F(x) - F(x-)$ by m.c.f.a. (Prop. 1.2.3; (2.4)), which is p_x on the support and 0 elsewhere. Density: $A \mapsto \int_A f\, dm$ is a measure (Cor. 2.3.6) that agrees with P on $\mathcal{C}$, so $P(A) = \int_A f\, dm$ on $\mathcal{B}(\mathbb{R})$ (Thm. 1.2.4).

Shortcut (2020 key). For a density f, set $P(A) \equiv \int_A f\,dm$ directly: P is a measure by Cor. 2.3.6, $P(\mathbb{R}) = 1$, and $X(\omega) = \omega$. Fine when the question says "argue"; 6410 asked for the Carathéodory steps.

Binomial$(10, 0.1)$ (6410 #2). $S = \{0, \dots, 10\}$, $F(x) = \sum_{k \in S,\, k \le x} \binom{10}{k}(0.1)^k(0.9)^{10-k}$ is constant on each $[j, j+1)$, and $F(\infty) = (0.1 + 0.9)^{10} = 1$. For (ii), with $A_k = X^{-1}(\{k\})$, $X^2 = \sum_{k=0}^{10} k^2 I_{A_k}$ a.e. (P) is simple, so

$$EX^2 = \sum_{k=0}^{10} k^2 \binom{10}{k}(0.1)^k(0.9)^{10-k} = np(1-p) + (np)^2 = 0.9 + 1 = 1.9$$

(Def. 2.3.2; Cor. 2.3.3 with Prop. 2.3.1(iii)).

Normal$(0, 1)$ (2020 #2). $F = \Phi$ is continuous, and μ_Φ is the $N(0,1)$ measure with $d\mu_\Phi/dm = \varphi$ (Ex. 4.1.1). $E\epsilon = \int x\varphi(x)\,dm = 0$ (given on the exam), so $EY_i = x_i\beta$ and $E\hat{\beta}_n = \beta$ by linearity (E8).

VII.3 Typical strategies for proofs

T1. Pointwise first. Decide convergence at each ω; name the exceptional set (often $\emptyset$ or one point) and show it is null (Defs. 2.5.1–2.5.2).

T2. Bound the bad set. $\{|f_n - f| > \epsilon\}$ lies inside the set where $f_n \neq f$; measure that set (Def. 2.5.3), or cite Thm. 2.5.1 (finite μ) or Markov (Thm. 2.5.3).

T3. Integrate simple functions exactly. For a spike $c_n I_{A_n}$, $\int |f_n - f|^p = c_n^p \mu(A_n)$ (Def. 2.3.2); find the threshold in p and check the boundary.

T4. UI by a witness, or by Vitali. For every t, exhibit n with $a_n(t) \ge c > 0$; or note that a.e. convergence, finite μ and no L^1 convergence rule out UI (Thm. 2.5.10, Remark 2.5.4).

T5. The standard machine. Indicators, then simple functions (linearity), then nonnegative functions (MCT), then integrable functions ($f^+ - f^-$) (Problems 2.39, 2.42).

T6. Subsequence principle. $x_n \to x$ iff every subsequence has a further subsequence converging to x (Problem 2.37(a)); with Thm. 2.5.2, every a.e. theorem becomes an in-measure theorem (E6).

T7. Fatou on a nonnegative difference. Apply Fatou to $g - f_n \ge 0$ or $g + f_n \ge 0$ and cancel a finite $\int g$ (DCT, EDCT, E1).

T8. Pick the convex function. For Jensen, name φ, verify convexity ($\varphi'' \ge 0$, Prop. 3.1.8), $P(X \in (a, b)) = 1$, and integrability of X and $\varphi(X)$ (Thm. 3.1.9).

T9. Factor 1 (or $|f|^q$). Write $1 = f^{1/2} \cdot f^{-1/2}$ for Cauchy–Schwarz, or $|f|^q = |f|^q \cdot 1$ for Hölder with exponent p/q (E2, E9).

T10. Case on finiteness. Treat $EX < \infty$ and $EX = \infty$ (or $\int g = \infty$) separately, with the conventions of $[0, \infty]$ (E10).

T11. Name the singular set. For $\perp$ and Lebesgue decompositions, exhibit B with $\mu_s(B^c) = 0$ and $\nu(B) = 0$, then split $A = (A \cap B) \sqcup (A \cap B^c)$ (E4, E5).

T12. Guess, then verify, a density. Propose h, check $\int_A h\,d\nu = \mu(A)$ for every A (Problem 2.39 helps), and quote uniqueness a.e. (Def. 4.1.3; E3, E11).

VII.4 Results proved on past exams

Result E1 (reverse Fatou; 2019 #2(a)). *If $f, f_n \in L^1$ and $f_n \le f$ for all n, then* $\limsup_n \int f_n\,d\mu \le \int \limsup_n f_n\,d\mu$.

Sketch. Fatou for the nonnegative $g_n = f - f_n$, then cancel the finite $\int f\,d\mu$ (T7).

Proof. $g_n = f - f_n \ge 0$ is measurable, so Fatou (Thm. 2.3.7) gives $\int \liminf g_n \le \liminf \int g_n$. Here $\liminf g_n = f - \limsup f_n$, and by linearity (Prop. 2.3.8), $\liminf \int g_n = \int f - \limsup \int f_n$. Since $(\limsup f_n)^+ \le f^+$, $\int \limsup f_n$ is defined in $[-\infty, \infty)$ and $\int (f - \limsup f_n) = \int f - \int \limsup f_n$. Hence $\int f - \int \limsup f_n \le \int f - \limsup \int f_n$; cancel $\int f \in \mathbb{R}$. //

Result E2 ($\int f \cdot \int f^{-1} \ge 1$; 2019 #2(b); Problem 3.8(a)). *Let μ be a probability measure, $f > 0$ a.e., and $f, 1/f \in L^1$. Then* $\big(\int f\,d\mu\big)\big(\int f^{-1}\,d\mu\big) \ge 1$.

Sketch. Split $1 = f^{1/2} \cdot f^{-1/2}$ and apply Cauchy–Schwarz (T9).

Proof. $f^{1/2}$ and $f^{-1/2}$ are in $L^2(\mu)$, since their squares are integrable. By Cor. 3.1.12,

$$1 = \mu(\Omega) = \int f^{1/2} f^{-1/2}\,d\mu \le \Big(\int f\,d\mu\Big)^{1/2}\Big(\int f^{-1}\,d\mu\Big)^{1/2};$$

square both sides. //

Note (slip in the problem). On a general measure space the same argument gives $\int f \cdot \int f^{-1} \ge \mu(\Omega)^2$, which can be below 1 ($\mu = \frac{1}{2}\delta_0$ and $f \equiv 1$ give $\frac{1}{4}$). The exam said "measure space"; the key's first step $1 = \int f^{1/2} f^{-1/2}\,d\mu$ uses $\mu(\Omega) = 1$, so state that assumption. Jensen with $\varphi(x) = 1/x$ on $(0, \infty)$ gives the same bound, $E(1/X) \ge 1/EX$.

Result E3 (equal integrals over every set; 2019 #2(c)). *If $f, g \in L^1$ and $\int_A f\,d\mu = \int_A g\,d\mu$ for all $A \in \mathcal{F}$, then $f = g$ a.e.*

Sketch. Test the hypothesis on $A = \{f > g\}$: a nonnegative function with integral 0 vanishes a.e.

Proof. $A = \{f > g\} \in \mathcal{F}$ (Prop. 2.1.3). By linearity (Prop. 2.3.8), $\int (f - g) I_A\,d\mu = 0$, and $(f - g)I_A \ge 0$, so $(f - g)I_A = 0$ a.e. by Prop. 2.3.9; that is, $\mu(f > g) = 0$. By symmetry, $\mu(g > f) = 0$. //

Apply. uniqueness of RN derivatives (Def. 4.1.3): for σ-finite μ and $f, g \ge 0$ that need not be integrable, run the same argument on $\{f > g\} \cap \{g \le k\} \cap \Omega_n$ with $\mu(\Omega_n) < \infty$. Problem 2.45 is the closed-set generalization.

Result E4 ($\ll$ and $\perp$ together force zero; 2019 #3(a), 6410 #3(c)). *If $\mu \ll \nu$ and $\mu \perp \nu$, then $\mu(A) = 0$ for every $A \in \mathcal{F}$.*

Sketch. The singular set B is μ-null, and B^c is ν-null, hence μ-null (T11).

Proof. Choose $B \in \mathcal{F}$ with $\mu(B) = 0$ and $\nu(B^c) = 0$ (Def. 4.1.2). Since $\mu \ll \nu$, $\mu(B^c) = 0$ (Def. 4.1.1). So $\mu(\Omega) = \mu(B) + \mu(B^c) = 0$, and $\mu(A) \le \mu(\Omega) = 0$ by monotonicity. //

The σ-finiteness assumed in 2019 is never used.

Result E5 (Lebesgue decomposition of a mixture; 2019 #3(b)–(c); Problem 4.5). *On $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ let $\mu_3 = \mu_1 + 3\mu_2$ with $\mu_1 = N(-1, 1)$ and $\mu_2 = Bernoulli(1/5)$, and let $\nu = m$. Then $\mu_{3a} = \mu_1$, $\mu_{3s} = 3\mu_2$,*

$$\frac{d\mu_{3a}}{dm}(x) = \varphi(x+1) = \frac{1}{\sqrt{2\pi}} e^{-(x+1)^2/2}, \qquad \text{and} \qquad \int x\,d\mu_{3a}(x) = -1.$$

Sketch. The density part is $\ll m$, the atoms sit on the m-null set $\{0,1\}$, and uniqueness of the decomposition identifies the two parts.

Proof. $\mu_1(A) = \int_A \varphi(x+1)\,dm$, so $\mu_1 \ll m$ (Ex. 4.1.3) with $d\mu_1/dm = \varphi(\cdot+1)$ (Def. 4.1.3). With $B = \{0,1\}$, $m(B) = 0$ and $3\mu_2(B^c) = 0$, so $3\mu_2 \perp m$ (Def. 4.1.2). Since μ_3 is finite and m is σ-finite, the decomposition in Thm. 4.1.1(i) is unique: $\mu_{3a} = \mu_1$ and $\mu_{3s} = 3\mu_2$. For (c), $\int |x|\varphi(x+1)\,dx < \infty$, so by Problem 2.39,

$$\int x\,d\mu_{3a}(x) = \int x\varphi(x+1)\,dm(x) = \int (y-1)\varphi(y)\,dm(y) = 0 - 1 = -1,$$

substituting $y = x+1$ (translation invariance, Problem 2.48(b)) and evaluating as limits of Riemann integrals over $[-n,n]$ with the DCT (Thm. 2.4.1, Problem 2.31). //

Note. $B = \mathbb{Z}_+$ works too (the key uses $\mathbb{N} \cup \{0\}$). The exam's part (c) says "as in (d)"; it means (b).

Result E6 (DCT under convergence in measure; 2020 #1(a)–(b); Problem 2.37(b)). *(a) DCT: if $|f_n| \le g$ a.e. for all n, $g \in L^1$, and $f_n \to f$ a.e., then $f \in L^1$, $\int f_n \to \int f$, and $\int |f_n - f| \to 0$ (Cor. 2.3.12). (b) The conclusion still holds when $f_n \to f$ in measure instead of a.e.*

Sketch. In-measure convergence gives a.e.-convergent sub-subsequences (Thm. 2.5.2), where the DCT applies; the subsequence principle (T6) lifts this to the whole sequence.

Proof of (b). Let $b_n = \int |f_n - f|\,d\mu$, and take any subsequence (n_k). Then $f_{n_k} \to f$ in measure, so some further subsequence $f_{n_{k_j}} \to f$ a.e. (Thm. 2.5.2). Then $|f| \le g$ a.e., so $f \in L^1$, and the DCT gives $b_{n_{k_j}} \to 0$. Every subsequence of (b_n) thus has a further subsequence tending to 0, so $b_n \to 0$ (Problem 2.37(a)), and $|\int f_n - \int f| \le b_n \to 0$. //

Result E7 (non-measurable functions; 2020 #1(c)–(d)). *Let $A \subset \mathbb{R}$ with $A \notin \mathcal{B}(\mathbb{R})$. Then (c) $f = I_A$ is not $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R})\rangle$-measurable, and (d) $f = I_A - I_{A^c}$ is not measurable, while $|f| \equiv 1$ is.*

Sketch. Pull back the Borel set $\{1\}$.

Proof. Non-Borel sets exist: a Vitali set is not even Lebesgue measurable, and the Cantor set has $2^{\mathfrak{c}}$ subsets while $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$ (Exam 1 Review, Section VI). In (c) and (d), $f^{-1}(\{1\}) = A \notin \mathcal{B}(\mathbb{R})$ although $\{1\} \in \mathcal{B}(\mathbb{R})$ (compare Ex. 2.1.3). In (d), $|f| \equiv 1$ is constant, hence measurable. //

Measurability of f gives that of $|f|$ (Props. 2.1.1(ii), 2.1.2), but not conversely.

Result E8 (a regression estimator; 2020 #2(b)–(c)). *Let $\epsilon_1, \ldots, \epsilon_n$ be $N(0,1)$ random variables, $\beta \in \mathbb{R}$, and $x_i \in \mathbb{R}$ with $\sum_i x_i^2 > 0$. Then $Y_i = x_i\beta + \epsilon_i$ and $\hat{\beta}_n = \sum_i x_i Y_i / \sum_i x_i^2$ are random variables, with $EY_i = x_i\beta$ and $E\hat{\beta}_n = \beta$.*

Proof. Constants are measurable, and sums and scalar multiples of measurable functions are measurable (Cor. 2.1.4), so Y_i and $\hat{\beta}_n$ are random variables. Each $\epsilon_i \in L^1$ with $E\epsilon_i = \int x\,d\mu_\Phi(x) = 0$ (given on the exam; change of variables, Problem 2.42). By linearity (Prop. 2.3.8), $EY_i = x_i\beta + E\epsilon_i = x_i\beta$ and $E\hat{\beta}_n = \sum_i x_i EY_i / \sum_i x_i^2 = \beta$. //

Result E9 ($L^p \Rightarrow L^q$ for $q < p$; 2020 #3(iv); Problem 3.17(a)). *On a probability space, if $0 < q < p < \infty$ and $X_n \to X$ in L^p, then $X_n \to X$ in L^q.*

Sketch. Lyapunov, $(E|Y|^q)^{1/q} \le (E|Y|^p)^{1/p}$, from Jensen with $\varphi(x) = |x|^{p/q}$.

Proof. Let $Y \in L^p$ and $r = p/q > 1$. Since $|Y|^q \le 1 + |Y|^p$, $|Y|^q \in L^1$. $\varphi(x) = |x|^r$ is convex on $\mathbb{R}$ (Ex. 3.1.1(a)) and $\varphi(|Y|^q) = |Y|^p \in L^1$, so Jensen (Thm. 3.1.9) gives $(E|Y|^q)^r \le E|Y|^p$. With

$Y = X_n - X$: $E|X_n - X|^q \le (E|X_n - X|^p)^{q/p} \to 0$; also $X_n, X \in L^q$ by the first bound, so Def. 2.5.4 holds. //

On a finite measure space use the Hölder form in Section V; when $\mu(\Omega) = \infty$ the implication fails (Section VIII).

Result E10 (a Jensen sandwich; 6410 #3(a); Problem 3.18(b)). *For a nonnegative random variable X, $\sqrt{1 + (EX)^2} \le E\sqrt{1 + X^2} \le 1 + EX$, including when $EX = \infty$.*

Sketch. $\varphi(x) = \sqrt{1 + x^2}$ is convex, and $x \le \varphi(x) \le 1 + x$ on $[0, \infty)$ supplies both the integrability Jensen needs and the infinite case (T8, T10).

Proof. $\varphi''(x) = (1 + x^2)^{-3/2} > 0$, so φ is convex (Prop. 3.1.8). If $EX < \infty$: $E\varphi(X) \le E(1 + X) = 1 + EX < \infty$ (Cor. 2.3.3), so Jensen (Thm. 3.1.9 with $(a, b) = \mathbb{R}$) gives $\varphi(EX) \le E\varphi(X)$. If $EX = \infty$: $E\varphi(X) \ge EX = \infty$, and all three quantities equal ∞. //

Result E11 (inverting a density; 6410 #3(b)). *If $\mu(A) = \int_A \varphi\, dm$ on $\mathcal{B}(\mathbb{R})$, then $\dfrac{dm}{d\mu}(x) = \sqrt{2\pi}\, e^{x^2/2}$.*

Sketch. $\mu \ll m$ with $d\mu/dm = \varphi > 0$, so invert (Prop. 4.1.2(iii)); then verify the defining identity directly (T12).

Proof. By Ex. 4.1.1, $\mu \ll m$ with $d\mu/dm = \varphi > 0$ everywhere; m and μ are σ-finite, so Prop. 4.1.2(iii) gives $m \ll \mu$ and $dm/d\mu = 1/\varphi$ a.e. (μ). Directly, Problem 2.39 with $h = I_A/\varphi$ gives $\int_A \varphi^{-1}\, d\mu = \int I_A \varphi^{-1} \varphi\, dm = m(A)$ for every $A \in \mathcal{B}(\mathbb{R})$. //

VII.5 Some potential next questions

Result E12 (Problem 3.18(a), the companion to 6410 #3(a)). *If $X > 0$ and EX, $E|\log X|$ and $E|X \log X|$ are finite, then $E(X \log X) \ge (EX)(E \log X)$.*

Sketch. Two Jensens: $x \log x$ is convex and log is concave on $(0, \infty)$.

Proof. $\varphi(x) = x \log x$ has $\varphi''(x) = 1/x > 0$ on $(0, \infty)$, so Jensen gives $E(X \log X) \ge (EX) \log(EX)$. Since $-\log$ is convex (Ex. 3.1.1(c)), Jensen gives $\log(EX) \ge E \log X$; multiply by $EX > 0$. //

Result E13 (Problem 3.8(b)). *If $f, g : [0, \infty) \to (0, \infty)$ are Borel with $f(x)g(x) \ge 1$ for all x, and $X \ge 0$, then $Ef(X)\, Eg(X) \ge 1$.*

Proof. Both expectations are positive. If one is infinite, there is nothing to prove. Otherwise $f(X)^{1/2}, g(X)^{1/2} \in L^2$, and Cauchy–Schwarz gives $1 \le E\sqrt{f(X)g(X)} \le (Ef(X))^{1/2}(Eg(X))^{1/2}$. //

With $f(x) = x$ and $g(x) = 1/x$ (for $X > 0$) this is E2. Problem 3.8(a) also asks about the correlation: $\mathrm{Cov}(X, 1/X) = 1 - EX \cdot E(1/X) \le 0$.

Result E14 (change of variables; Problem 2.42, "His favorite"). *Let $f : \Omega_1 \to \Omega_2$ be $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$-measurable, $\mu_2 = \mu_1 f^{-1}$, and $h : \Omega_2 \to \mathbb{R}$ be $\langle \mathcal{F}_2, \mathcal{B}(\mathbb{R}) \rangle$-measurable. Then $h \circ f \in L^1(\mu_1)$ iff $h \in L^1(\mu_2)$, and $\int h \circ f\, d\mu_1 = \int h\, d\mu_2$.*

Sketch. The standard machine (T5).

Proof. For $h = I_A$ with $A \in \mathcal{F}_2$: $\int I_A \circ f\, d\mu_1 = \mu_1(f^{-1}(A)) = \mu_2(A) = \int I_A\, d\mu_2$. By linearity the identity holds for nonnegative simple h. For measurable $h \ge 0$, take simple $0 \le h_k \uparrow h$; then $h_k \circ f \uparrow h \circ f$, and the MCT on both sides gives the identity. Applied to $|h|$, this gives the

integrability statement; applied to $h^{\pm}$ (note $(h \circ f)^{\pm} = h^{\pm} \circ f$), it gives the identity for integrable h. //

Apply. $Eh(X) = \int h \, dP_X$, e.g. $E\epsilon = \int x \, d\mu_\Phi(x)$ in E8. The problem's third integral, $\int x \, d\mu_3(x)$ with $\mu_3 = \mu_2 h^{-1}$, is the same result applied to h and the identity map.

Result E15 (L^1 convergence gives UI; Problem 2.36(a)). *If $f_n \to 0$ in $L^1(\mu)$, then $\{f_n\}$ is UI.*

Sketch. Finitely many functions are UI; the tail is small in L^1.

Proof. Fix $\epsilon > 0$ and choose N with $\int |f_n| \, d\mu < \epsilon$ for $n > N$; then $a_n(t) \le \int |f_n| \, d\mu < \epsilon$ for all $n > N$ and all t. The finite family $\{f_1, \ldots, f_N\}$ is UI (Prop. 2.5.7(i)), so some t_ϵ has $a_n(t) < \epsilon$ for all $n \le N$ and $t \ge t_\epsilon$. Hence $\sup_n a_n(t) \le \epsilon$ for $t \ge t_\epsilon$. //

With Prop. 2.5.7(iv), $f_n \to f$ in L^1 also gives UI. Together with Thm. 2.5.10: on a finite measure space with $f_n \to f$ a.e., UI holds iff the convergence is in L^1.

Result E16 (in measure, convergent nowhere; Problem 2.46). *List $g_{m,k} = I_{[(k-1)/m,\, k/m]}$, $1 \le k \le m$, $m \ge 1$, as $f_1, f_2, \ldots$ on $([0,1], \mathcal{B}([0,1]), m)$. Then $\int f_n \, dm \to 0$, so $f_n \to 0$ in L^p and in measure, but $f_n(x)$ converges for no x.*

Proof. If $f_n = g_{m,k}$, then $\int f_n \, dm = 1/m$, and $m \to \infty$ as $n \to \infty$. Fix x. For each $m \ge 3$, some $g_{m,k}$ equals 1 at x and some $g_{m,k'}$ equals 0 at x, so $f_n(x) = 1$ and $f_n(x) = 0$ each happen infinitely often. //

Thm. 2.5.2 only promises a subsequence: here $g_{m,1} = I_{[0,1/m]} \to 0$ on $(0,1]$.

Result E17 (more decompositions; Problem 4.5, with Tables 4.6.1–4.6.2). Find each measure's support, divide densities on the common part (Prop. 4.1.2(i), (iii)), and call the rest singular.

- (a) $\mu = N(0,1)$, $\nu = \text{Exponential}(1)$: $\mu_a(A) = \mu(A \cap (0,\infty))$ with $\frac{d\mu_a}{d\nu}(x) = \varphi(x) e^x I_{(0,\infty)}(x)$, and $\mu_s(A) = \mu(A \cap (-\infty, 0])$, singular set $B = (-\infty, 0]$.
- (b) $\mu = \text{Exponential}(1)$, $\nu = N(0,1)$: $\mu \ll \nu$ with $\frac{d\mu}{d\nu}(x) = \sqrt{2\pi}\, e^{x^2/2 - x} I_{(0,\infty)}(x)$.
- (c) $\mu = N(0,1) + \text{Poisson}(1)$, $\nu = \text{Cauchy}(0,1)$: $\mu_a = N(0,1)$ with $\frac{d\mu_a}{d\nu}(x) = \pi(1+x^2)\varphi(x)$, and $\mu_s = \text{Poisson}(1)$, with $B = \mathbb{Z}_+$.
- (f) $\mu = \text{Binomial}(10, 1/2)$, $\nu = \text{Poisson}(1)$: $\mu \ll \nu$ with $\frac{d\mu}{d\nu}(i) = \binom{10}{i} 2^{-10} e\, i!$ for $i = 0, \ldots, 10$, and 0 elsewhere.

Also worth a look. Problems 2.28–2.30 (the UI examples in Section VIII); 2.43–2.44 (Scheffé); 2.45; 3.13 (Borel–Cantelli); 3.16; 3.21 (for which p is $x^2 \in L^p(\mu_F)$? All $p > 0$ for Φ; $0 < p < \frac{1}{2}$ for the Cauchy cdf); 4.3 (why σ-finiteness); 4.7(c) (Binomial to Poisson in total variation); 4.10; 4.13(b).

VIII Examples, counterexamples and a self-check

Phenomenon	Example	Where
a.e., not in measure ($\mu(\Omega) = \infty$)	$I_{[n,\infty)}$ on $(\mathbb{R}, m)$	after Thm. 2.5.1
In measure and L^p, convergent nowhere	typewriter $g_{m,k}$	Prob. 2.46, E16
a.e. and in measure, not L^1	$nI_{[0,1/n]}$ on $[0,1]$	after Thm. 2.5.3; 6410 #1
L^1-bounded, not UI	$nI_{[0,1/n]}$ on $[0,1]$	Remark 2.5.3
a.e. and $\int f_n \to \int f$, not UI	$nI_{(0,1/n)} - nI_{(-1/n,0)}$ on $[-1,1]$	Prob. 2.28
a.e. and UI, $\int f_n \not\to \int f$ ($\mu(\Omega) = \infty$)	$n^{-1/2} I_{(0,n)}$ on $(\mathbb{R}, m)$	Prob. 2.30
Strict inequality in Fatou	$I_{[n,\infty)}$ on $\mathbb{R}$; $nI_{[0,1/n]}$ on $[0,1]$	Remark 2.3.6
Nearly uniform, not L^∞	$nI_{[0,1/n]}$ on $[0,1]$	after Def. 2.5.7
Lebesgue, not Riemann integrable	$I_{\mathbb{Q}\cap[0,1]}$	Prob. 2.25
f not measurable, $\lvert f\rvert$ measurable	$I_A - I_{A^c}$, $A \notin \mathcal{B}(\mathbb{R})$	2020 #1(d), E7
$L^2 \not\subset L^1$ when $\mu(\Omega) = \infty$	$1/x$ on $(1,\infty)$	§3.2
$L^1 \not\subset L^2$, even on $[0,1]$	$x^{-1/2}$ on $(0,1)$	§3.2
Triangle inequality fails for $(\int \lvert f\rvert^p)^{1/p}$, $p < 1$	$I_{[0,1/2]}$, $I_{(1/2,1]}$ with $p = \frac{1}{2}$: $1 > \frac{1}{4} + \frac{1}{4}$	Prob. 3.12(a)
Radon–Nikodym fails without σ-finiteness	$m \ll$ counting measure on $[0,1]$	Remark 4.1.1, Prob. 4.3
Mutually singular	m and counting on $\mathbb{Z}$; $m\lvert_{(-\infty,0]}$ and Exp(1)	Exs. 4.1.4–4.1.5
Mutually absolutely continuous	Poisson(λ) and counting on $\mathbb{Z}_+$; $N(0,1)$ and m	Exs. 4.1.1–4.1.2
Continuous cdf singular to m; continuous but not a.c.	the Cantor function	§4.5.3
a.c. on bounded intervals, not on $\mathbb{R}$	$F(x) = x^2$	§4.4
Continuous, not BV	$x^2 \sin(1/x^2)$ on $[0,1]$	Remark 4.3.2(e)

Self-check

Question	Answer
a.e. $\Rightarrow$ in measure?	Yes if $\mu(\Omega) < \infty$ (Thm. 2.5.1); not in general ($I_{[n,\infty)}$).
In measure $\Rightarrow$ a.e.?	Only along a subsequence (Thm. 2.5.2); the typewriter converges nowhere (E16).
$L^p \Rightarrow$ in measure? The converse?	Yes, by Markov (Thm. 2.5.3). No ($nI_{[0,1/n]}$); yes with domination (DCT) or with $\{\|f_n\|^p\}$ UI and $\mu(\Omega) < \infty$ (Prob. 2.36(b)).
Hypotheses of the DCT?	$\|f_n\| \le g$ a.e. for all n, $g \in L^1$, and $f_n \to f$ a.e. (or in measure, E6).
When does UI give L^1 convergence?	$\mu(\Omega) < \infty$ plus a.e. (or in-measure) convergence (Thm. 2.5.10; Prob. 2.29(b)).
Fastest "not UI" for $f_n \ge 0$, $f_n \to f$ a.e., $\mu(\Omega) < \infty$?	$\int f_n \not\to \int f$ (Remark 2.5.4). For a spike, $a_n(t) = c_n\mu(A_n) \not\to 0$.
Is an L^1-bounded family UI?	Not necessarily ($nI_{[0,1/n]}$); $\sup_\lambda \int \|f_\lambda\|^{1+\epsilon} < \infty$ suffices (Prop. 2.5.7(ii)).
Hypotheses of Jensen?	Probability measure; $P(X \in (a,b)) = 1$; φ convex on (a,b); $E\|X\| < \infty$ and $E\|\varphi(X)\| < \infty$.
When is Jensen an equality?	For strictly convex φ, iff X is a.s. constant (Remark 3.1.3).
$L^p \subset L^q$?	For $q < p$ when $\mu(\Omega) < \infty$ (Hölder); never automatic when $\mu(\Omega) = \infty$.
Is $\|\cdot\|_p$ a norm? Is L^p complete?	A norm for $1 \le p \le \infty$ (Minkowski); for $0 < p < 1$ use $d_p = \int \|f-g\|^p$ (Remark 3.2.1). Complete for all $0 < p \le \infty$ (Thm. 3.2.2).
Hypotheses of Radon–Nikodym?	$\mu \ll \nu$ with μ, ν σ-finite (Thm. 4.1.1); without σ-finiteness it can fail (Remark 4.1.1).
Is $d\mu/d\nu$ unique?	Up to ν-null sets, for σ-finite μ (Def. 4.1.3; E3).
$d\nu/d\mu$ from $d\mu/d\nu$?	$(d\mu/d\nu)^{-1}$ a.e. (μ), when $d\mu/d\nu > 0$ a.e. (ν) (Prop. 4.1.2(iii); E11).
$\mu \ll \nu$ and $\mu \perp \nu$?	$\mu = 0$ (E4).
Decomposing "density plus atoms" against m?	The density part is μ_a; the atoms are μ_s, with singular set their locations (E5).
What is a finite signed measure?	A difference of finite measures, or $\int_A f\,d\mu$ with $f \in L^1(\mu)$ (Thm. 4.2.2); split by Hahn and Jordan (Thm. 4.2.3).
How to compute $Eh(X)$?	$\int h\,dP_X$ (Prob. 2.42); with a density, $\int hf\,dm$ (Prob. 2.39); then Riemann (Thm. 2.4.1).

IX Appendix: tools from analysis and probability

- **Subsequence criterion** (Problem 2.37(a)). $x_n \to x$ in $\bar{\mathbb{R}}$ iff every subsequence has a further subsequence converging to x.
- **liminf and limsup.** $\liminf(c - a_n) = c - \limsup a_n$; $\liminf a_n + \liminf b_n \le \liminf(a_n + b_n)$; $x_n \to x$ iff $\liminf x_n = \limsup x_n = x$.
- **Series and limits.** $\sum n^{-p} < \infty$ iff $p > 1$ (Problem 3.19); $\sum_{j \ge 0} r^j = 1/(1-r)$ for $0 < r < 1$; $(1 + x_n/n)^n \to e^x$ when $x_n \to x$ (Problem 2.35); $n^a \to 0$ iff $a < 0$.
- **Borel–Cantelli** (Problem 3.13). $\sum_k \mu(A_k) < \infty \Rightarrow \mu(\limsup_k A_k) = 0$.
- **Elementary inequalities.** $(x+y)^p \le x^p + y^p$ for $x, y \ge 0$, $0 < p \le 1$ (2.2); $|a+b|^p \le 2^{p-1}(|a|^p + |b|^p)$ for $p \ge 1$ (1.12); Young, $ab \le a^p/p + b^q/q$ (from (1.10)); $e^x \ge 1 + x$; $\log x \le x - 1$; $x \le \sqrt{1+x^2} \le 1 + x$ for $x \ge 0$.
- **Second derivatives for convexity** (Prop. 3.1.8). $(\sqrt{1+x^2})'' = (1+x^2)^{-3/2}$; $(x \log x)'' = 1/x$; $(1/x)'' = 2/x^3$; $(x^r)'' = r(r-1)x^{r-2}$, which is ≥ 0 on $(0, \infty)$ for $r \ge 1$.
- **Moments.** Binomial(n,p): mean np, variance $np(1-p)$, $EX^2 = np(1-p) + (np)^2$; Poisson(λ):

mean and variance λ; Exponential(β): mean β, variance β^2; $N(\gamma, \sigma^2)$: mean γ, variance σ^2.

- **Gaussian integrals.** $\int e^{-x^2/2}\,dx = \sqrt{2\pi}$; $\int x\varphi(x)\,dx = 0$; $\int x^2\varphi(x)\,dx = 1$; the $N(\gamma, \sigma^2)$ density is $\sigma^{-1}\varphi((x-\gamma)/\sigma)$.
- **Invariance of** m (Problem 2.48). $m(E + c) = m(-E) = m(E)$, and $\int f(x + c)\,dm = \int f(-x)\,dm = \int f\,dm$ for $f \in L^1(m)$.
- **Non-measurable sets** (§1.3.2(v); Exam 1 Review, Section VI). A Vitali set is not Lebesgue measurable; the Cantor set has $2^{\mathfrak{c}}$ subsets but $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$, so some Lebesgue measurable sets are not Borel.
- **Heine–Borel** (Thm. A.4.2). $K \subset \mathbb{R}$ is compact iff it is closed and bounded (used for countable additivity of μ_F on $\mathcal{C}$).